

KEVIN SCHWARZWALD

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Interdisciplinary climate scientist helping inform climate impacts projections and decision-making under uncertainty

EDUCATION

Columbia University

Doctor of Philosophy, Earth and Environmental Sciences

New York, NY

October 2024

Master of Philosophy, Earth and Environmental Sciences

October 2023

Master of Arts, Earth and Environmental Sciences

May 2021

Research Areas: climate variability, uncertainty in projections of climate and climate impacts, precipitation in East Africa, tropical convection, climate model evaluation, the use of climate projections in policy and adaptation efforts

Advisors: Lisa Goddard (†2022), Richard Seager; *Committee Members*: Mingfang Ting, Kate Marvel, Radley Horton

- *PhD thesis*: *The Present and Future of the Horn of Africa Rains*

Peking University, Yenching Academy (北京大学燕京学堂)

Beijing, China

Master of Law in China Studies (concentration in Politics and International Relations)

July 2017

Honors: Awards for Outstanding Academic Achievement and Outstanding Contributions to the Academy

- *Master's thesis*: *The Relationship between Fiscal Systems and Inefficient Urban Expansion in the PRC: An Empirical Look at Prefecture-Level Urbanization*

The University of Chicago

Chicago, IL

Bachelor of the Arts (A.B.) in Physics; Public Policy (concentration in Economics)

June 2015

Honors: Graduated with General Honors, Dean's List 2011-2013, 2015, Departmental Honors in Public Policy

- *Senior Thesis (Public Policy)*: *"The Wrong Folks Kept Winning": Protectionism and America's Trains*
- Study Abroad: Paris, France

ACADEMIC RESEARCH EXPERIENCE

Columbia Climate School

September 2024 - present

Climate School Postdoctoral Scientist

New York, NY

Mentors: Radley Horton and Gernot Wagner

- Current projects include: understanding and constraining the range of climate uncertainty in projections of climate impacts; the suitability of Global Warming Levels in regional climate impacts studies; improving forecast-based financing methodologies for anticipatory humanitarian actions

International Research Institute for Climate and Society (IRI),

September 2019 – August 2024

Lamont-Doherty Earth Observatory (LDEO)

Graduate Research Assistant

New York, NY

- Projects included: the present and future of rainfall dynamics in East Africa, understanding the sources of climate uncertainty in projections of the societal impacts (agriculture, mortality, etc.) of climate change, and building tutorials and python tools to make climate data more accessible to end users

Energy Policy Institute at the University of Chicago (EPIC)

September 2017 – July 2019

Pre-Doctoral Fellow (joint with RDCEP)

Chicago, IL and remotely from New York, NY as GISS Visiting Researcher

Principal Investigator: Amir Jina

- Projects included: adapting latest downscaling and bias-correction methods to impacts studies, understanding how changes in climate variability impacts common economic damage
- Advised on the use of climate model and observational data in economics and policy applications

Center for Robust Decision-Making on Climate and Energy Policy (RDCEP)

April 2015 – July 2019

Pre-Doctoral Fellow (joint with EPIC 2017-2019)

Chicago, IL and remotely from New York, NY, and Beijing, China

Principal Investigator: Elisabeth Moyer

- Projects included: understanding changes in precipitation and temperature variability from synoptic to inter-decadal timescales in global climate models

PUBLICATIONS (upcoming)

Schwarzwald, Kevin, Nathan Lenssen, Radley Horton, and Gernot Wagner. "A bias-corrected & downscaled massive ensemble to diagnose uncertainty in climate impact projections." *Under review at Nature Scientific Data*. [Link](#).

Mauerman, Max, **Kevin Schwarzwald**, Daniel Osgood, Nathan Lenssen, and Abdel-Lathif Younous. “Sampling Uncertainty is Critical for Setting Forecast-Based Anticipatory Action Triggers” *Under revision at Bulletin of the American Meteorological Society*.

Lenoble, Colin, Mathias Valla, Jonas Schmitt, Céline Guivarch, Améline Vallet, Pauline Rivoire, **Kevin Schwarzwald**, and Kai Kornhuber. “Global risks of renewable energy droughts under climate change” *Under review at Nature Communications*.

Schwarzwald, Kevin and Nathan Lenssen. *Solicited textbook chapter, in progress*. “The Sources of Uncertainty in Climate Projections.” in *Encyclopedia of Climate System Science*, edited by Sumant Nigam and Isla R. Simpson. Elsevier Major Reference Works. Elsevier.

PUBLICATIONS (peer-reviewed)

Schwarzwald, Kevin and Kerrie Geil. 2024. “xagg: A Python Package to Aggregate Gridded Data onto Polygons.” *Journal of Open Source Software* 9 (104): 7239. [Link](#).

Schwarzwald, Kevin and Richard Seager. 2024. “Revisiting the ‘East African Paradox’: CMIP6 models also struggle to reproduce strong observed MAM drying trends.” *Journal of Climate* 37 (24): 6641–58. [Link](#).

Rising, James A., Azhar Hussain, **Kevin Schwarzwald**, and Ana Trisovic. 2024. “A Practical Guide to Climate Econometrics: Navigating Key Decision Points in Weather and Climate Data Analysis.” *Journal of Open Source Education* 7 (75): 90 [Link](#).

Schwarzwald, Kevin, Richard Seager, Mingfang Ting, and Alessandra Giannini. 2023. “Large-Scale Stability and the Greater Horn of Africa Long and Short Rains.” *Journal of Climate* 36 (20): 7297–7317. [Link](#).

Anderson, Weston, Benjamin I. Cook, Kim Slinski, **Kevin Schwarzwald**, Amy McNally, and Chris Funk. 2023. “Multiyear La Niña Events and Multiseason Drought in the Horn of Africa.” *Journal of Hydrometeorology* 24 (1): 119–31. [Link](#).

Schwarzwald, Kevin, and Nathan Lenssen. 2022. “The Importance of Internal Climate Variability in Climate Impact Projections.” *Proceedings of the National Academy of Sciences* 119 (42). [Link](#).

Schwarzwald, Kevin, Lisa Goddard, Richard Seager, Mingfang Ting, and Kate Marvel. 2022. “Understanding CMIP6 Biases in the Representation of the Greater Horn of Africa Long and Short Rains.” *Climate Dynamics* 61 (3): 1229–55. [Link](#).

Schwarzwald, Kevin, Andrew Poppick, Maria Rugenstein, Jonah Bloch-Johnson, Jiali Wang, David McInerney, and Elisabeth J. Moyer. 2021. “Changes in Future Precipitation Mean and Variability across Scales.” *Journal of Climate* 34 (7): 2741–58. [Link](#).

AWARDS, GRANTS, COMPETITIVE PROGRAMS

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| - Climate School Postdoctoral Program; successful institutional fellowship application for “ <i>The Full Magnitude of Climate Uncertainty in Projections of Climate Impacts: A Case Study in Tropical, Developing Countries</i> ”, \$145,430 (salary) | 2024-2026 |
| - CATER School on Transdisciplinary Climate Risk Adaptation and Action (Naivasha, Kenya; fully funded, including travel) | 2023 |
| - ASciNA Young Scientist Award, \$7500 | 2023 |
| - Outstanding Oral Presentation Award, 103 rd AMS Annual Meeting | 2023 |
| - Center for Science and Society Seed Grants, \$3975 total (to NENSIC, co-PI) | 2021, 2022, 2023 |
| - ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics (Trieste, Italy; fully funded) | 2022 |
| - Yenching Academy Dean’s Research Grant, ¥5000 | 2016 |

OTHER EXPERIENCE

New Era Network for Societally Integrated Climatology (NENSIC)

May 2020 – present

Co-founder, Organizer

Website: nensic.org

- Launched an interdisciplinary network designed to connect young researchers, students, and professionals working on climate issues in any field, with a focus on improving interdisciplinary communication, building informal connections, peer mentoring, and providing a welcoming, inclusive venue to present work and find collaborators
- Organized and co-organized 60+ events in person and remotely with participants from four continents, with research talks on topics such as international climate negotiations, writing public comments as a scientist, or the coordination of disaster planning in Ecuador, in addition to networking events, tutorials, and interdisciplinary job panels

Eparque Urban Strategies <i>Independent Consultant (part-time)</i>	March 2019 – July 2019 New York, NY
Yenching Global Symposium (全球青年中国论坛) <i>Associate Director, Logistics</i>	October 2015 – April 2016 Beijing, China
Technology Policy Institute <i>Google Policy Fellow</i>	July 2014 – August 2014 Washington, DC
Paris Center for Cosmological Physics <i>Lab Intern</i>	January 2014 – March 2014 Paris, France
alpS GmbH <i>Management Intern, Consulting Group</i>	August 2013 – September 2013 Innsbruck, Austria
Moser/Jaritz <i>Research Intern</i>	August 2013 – September 2013 Gmunden, Austria

INVITED AND CONTRIBUTED PRESENTATIONS

2026: International Institute for Applied Systems Analysis (IIASA), Mailman School of Public Health, EGU General Assembly (**highlighted talk**)

2025: MLISTRAL2 Meeting (**invited keynote**), American Geophysical Union (AGU) Fall Meeting

2024: JPL Science Understanding through Data Science (SUDS) Seminar; European Geosciences Union (EGU)'s General Assembly, Max Planck Institute for Meteorology Hamburg, Oxford School of Geography and the Environment

2023: American Meteorological Association (AMS)'s Annual Meeting; EGU General Assembly; University of California Santa Barbara (UCSB) Climate Hazards Center; ECS Symposium; WCRP Open Science Conference; AGU Fall Meeting (**invited abstract**)

2022: AGU Fall Meeting; Institute for Mathematical and Statistical Innovation (IMSI); NASA Goddard Institute for Space Studies (GISS); ICTP Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics; Interdisciplinary PhD Student Workshop in Sustainable Development (IPWSD); Columbia Student Climate Symposium

2021: AGU Fall Meeting; University of Chicago Environmental Data Science Lunch

2019: US CLIVAR Large Ensembles Workshop

2018: AGU Fall Meeting

DATA TOOLS AND TUTORIALS

Weather Panel Tutorial

climateestimate.net

- In an inter-disciplinary team, co-developed a tutorial designed to introduce climate economists to working with observational climate data using python (xarray), R, or MATLAB

xagg

github.com/ks905383/xagg

- Lead developer and maintainer of **xagg**, a python package for aggregating raster data (such as gridded weather data) onto polygons (such as administrative shapefiles), using area-averaging based on the overlap between pixels, polygons

quantproj

github.com/ks905383/quantproj

- Developer of **quantproj**, an R package for projecting weather data by using downscaled fine-scaled distributional changes in Large Ensembles and the delta method; based on the methodology of Haugen et al., 2018
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SKILLS

Analysis: advanced proficiency in python (xarray, dask), MATLAB, R, and Excel, basic proficiency in GIS (QGIS) and Stata, experience with climate (NetCDF, zarr, nco, CMIP, etc.) and satellite data analysis (DMSP/OLS, VIIRS, etc.)

High-performance computing: proficiency with work on HPC systems, including working with schedulers, writing optimized and parallelized code, etc., proficiency with NCAR's HPC and file storage system

Modeling: experience in running SPEEDY, an intermediate-complexity global climate model (GCM)

Communication/Design: advanced proficiency in LaTeX, PowerPoint, Keynote, inkscape; basic proficiency in Adobe Illustrator

Language: native in German, fluent in French, basic communication skills in Mandarin Chinese

PROFESSIONAL NETWORKS

- American Geophysical Union (AGU)
- European Geosciences Union (EGU)
- Austrian Scientists & Scholars in North America (ASciNA)

